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Python Assignment 1

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① # age-input.py

while True :

try :

age = int (input (Enter your age :))

break

except ValueError :

print ("Invalid Input, Please enter a whole number for your age")

print (f Your age is {age}.)

② # fruits.py

fruits = [Apple, Banana, Orange, Mango, Grapes]

Write the fruits to fruits.txt one per line

with open (fruit.txt, w) as file :

for fruit in fruits :

file.write (fruit + \n)

print (Fruits saved to fruits.txt)

Read the file & display each fruit print (\n

your favourite fruits are: ")

with open('fruits.txt', 'r') as file:
for line in file:

print(line.strip())

③ students = {

Billy : 88

David : 75

Teddy : 92

Melissa : 64

Carol : 79

print(student and their marks:)

for name, mark in student.items():

print(f'{name}: {mark}')

average = sum(student.values()) / len(student)

print(f'\n Average mark : {average : 2f}')

max.mark = max(student.values())

min.mark = min(student.values())

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```
top_students = [n for n, m in students.items() if  
m = max_mark]
```

```
bottom_students = [n for n, m in students.items() if  
m = min_mark]
```

```
print("Students sorted by marks (highest to lowest):")  
for name, mark in sorted(students.items(), key=lambda  
item: item[1], reverse=True):
```

```
    print(f"{name}: {mark}")
```

```
while True:
```

```
    try
```

```
        threshold = int(input("Enter the pass threshold  
(0-100): "))
```

```
        if not (0 <= threshold <= 100):
```

```
            print("Please enter a number between 0 and 100")
```

```
            continue
```

```
        break
```

```
    except ValueError
```

```
        print("Invalid input. Please enter a whole number  
for the threshold")
```

```
    print(f"Students who scored at least {threshold}:")
```

```
    if passed
```

```
        for name in passed
```

```
print(-, name  
else  
print (No students passed
```

```
# book_demo.py
```

```
class book:
```

```
def __init__(self, title: str, author: str, price: float):
```

```
    self.title = title
```

```
    self.author = author
```

```
    self.price = price
```

```
def display(self) -> None:
```

```
    Prints the details of the book
```

```
    print(f Title : {self.title})
```

```
    print(f Author : {self.author})
```

```
    print(f Price : {self.price})
```

```
book 1 = Book ( 1984, George Orwell, 9.99)
```

```
book 2 = Book ( To Kill a Mockingbird, Harper Lee, 12.50)
```

```
book 1. display()
```

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print()

book 2 . display()

print()